

ActInflab ~ 2022 Quarterly Roundtable #2

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[Music] all right hello welcome everyone this is the actin flap quarterly round table number 20. number two in 2022 it's june 29th so welcome to the active inference lab we're a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links on this slide this is recorded in an archive live stream so please provide us with feedback so we can improve our work all backgrounds and perspectives are welcome and we'll be following video etiquette for live streams today it's a quarterly roundtable lab meeting so we're going to have a lot of updates to share on some of the activities of the second quarter of 2022 and also share a lot of next steps at the project and at the organizational scale so let's begin with an introduction slash warm-up we can say hello and maybe as if it were a paper we were discussing what was just one memory or something that you learned this quarter or some participation that you just want to call out and highlight so i'll go first i'm daniel i'm a researcher in california and have a lot to add but just saw so much amazing participation and development this quarter um like without having compiled these slides with everybody so much would have just um it all happened so fast and so looking forward to catching up on this and also building structures so that we can even have more clarity internally and externally about what we're working on like there's just quite a contrast with hopefully this lab meeting and even our previous five or six and i'll pass it to blue yeah this quarter was awesome i think the launch of the textbook group has been like pretty mind-blowing for me um that's like a new uh thing that we've done and um yeah i am enjoying it and looking forward to future cohorts and part two of the textbook group but that to me was kind of um like like a pivotal a pivotal moment in this last three months uh and i'll pass it to jakob so hi i'm markup i'm a student in the uk the textbook group was also the highlight uh from this quarter for me um and i one thing that i learned probably would have to be uh that multi-agent active inference is a lot more interesting and harder than initially meets the eye and i'll pass it to alex yeah thanks i'm alex and i can agree that education as a survivor from the lab is the most important thing for this quarter so for study group as well as for our farmers what we are planning and now preparing for participants and more broader community is the

most important thing from our deposit to iran thank you alex my name is ivan i'm in romania i'm a researcher in systems management school last quarter we with our team done a lot of work uh in case of education as well as in case of administration in a lab in the lab level so i'm looking forward to development for all the directions we are we are going to all right let's get into the projects so first we're going to provide some updates on the project scale so these are things that happen during second quarter for a variety of projects all right first as mentioned by several of us here we began the first cohort of the active inference textbook group in may 2022 and that'll go through july 2022 and we are working on the first five chapters of this active inference textbook by thomas parr giovanni pizzulo and carl fristen and just to give a little taste though there's a few other pieces too that people who participate in future cohorts will get to experience um there's a collaborative questions and discourse affordance which has allowed people to introduce basic clarifying questions as well as questions about application and about philosophy all different kinds of questions have been addressed in a way that really clearly references which part of the text it's about which page or which figure and in fact all the tables figures equations and so on have been annotated using coda such that by just clicking on it you can see a pop-up of a figure and that allows a very interactive way of dealing with different terms and different objects of attention and then on the right we have some of the work by those engaged most in the math learning where for each equation we are aiming to have a natural language description first just reading out the equation as it is and also here active inference ontology terms are in blue so this will facilitate the translation of equations across human languages as well as the development of multiple programming languages maybe that are representing a certain equation and giving a lot of continuity with the terms and the tools and the points of reference within the book and all of that is being threaded together really by the active inference ontology which has been in development since really the initial days of the lab so it's just awesome to see this new knowledge artifact come on to the scene a new modification in the niche new textbook has entered the chat and then be able to organize rapidly and have a bunch of participants get involved um what would anybody else like to add about the textbook group it's different than a paper it has similarities with papers but it is very different so it's been a great experience anything else though yeah alex yeah i think it was also interesting interesting experience how subgroup for math learning was organized like by initiative of participants and how we started to work on like additional direction and provide more energy and more efforts to structure math side of the book what yeah and this is our first attempt in such activity and as i see it it has a lot of opportunities for learners really helps to

acknowledge all the concepts and all the idea of active inference great okay um moving on to another educational effort the internal systems thinking ongoing activity alex would you like to describe a little bit here or on i can say that from the beginning it was part of our strategy to use system thinking intelligent to organize work on projects and it is like a requirement for lab participants to have an understanding of concepts of system thinking so having this educational activity for people who are willing to learn it and later to take part in different roles in labs project for for us it's like a basement for all we're doing and we are also learning by doing this education and hopefully new version of textbook and the program which we're using here will help us to be more and more efficient in all projects that we are doing any other comments here yeah we launched this type of activity to have a common language for all of the participants and this course is providing such language it is uh one of the uh one of the such activity and i hope we will continue with different different types of educational formats great continuing on we have also in education there was a lot of development in q2 with our knowledge engineering and knowledge corpus project blue would you like to provide initial summary sure so i think we've described this project before but just briefly we are taking the open source papers from pubmed that have active inference or free energy principle as keywords and we've put all of these papers into a database and with all of the authors and all of the metadata and and we are then scraping them and parsing out the text for keywords from our ontology and this is shown here in the heat map this top right figure it shows all the different like ontology words and how they they cluster per paper so we've looked at keyword clustering and authorship like this is first author papers by year we can kind of see trends in active inference but but this will serve as then a way for our ontology to access the um original data so if you're looking for words in the ontology something you don't understand like markov blanket or something like this you can actually find go straight to the horse's mouth so to speak and find papers that reference this maybe that reference it a lot and so we're intending to publish this and then really we want to build a reflexive system and invite people to upload their own papers and we're also starting to get papers from archive now um and then we also will invite people to upload papers that maybe are not open access that they want to have catalogued in our database so it's really a fun interesting data sciencey project that i've enjoyed working on with daniel and our colleague rj so yeah yeah just just to um show that that heat map again where the columns are the terms and the frequency of their use and the rows are different clusters of papers so this is like topic modeling it's identifying clusters of terms that have a similar coexpression pattern and so it was really fun to apply this kind of uh heat

mapping and clustering and some statistical techniques because both blue and i have worked with gene expression analyses where the rows would be like the genes in the genome and the columns would be how much different genes are expressed in different samples and so the ontology is kind of like the genome because there's one copy of every term it's like a dictionary and then some terms are used a lot or little or not at all in any given library in any given sample of text and so it's showing how the expression of the ontology and of terms relates to this in our case a reference genome of not just the whole language dictionary but specifically the active ontology as a versioning and developing artifact so then rhetorical and conceptual relationships can be drawn in principle amongst ontology terms like here's the relationship between markov blanket and frist and blanket or whatever other terms are being linked and also that can be integrated with empirical results from the literature and also translated across different language to facilitate all kinds of and research assistance so this will be really fun and as blue mentioned we're hoping to release the written paper and the tooling around this kind of a pipeline which includes people being able to submit papers in annotate and update the information and then also have a rendering output hopefully we'll provide an update on that soon but it was a fun quarter for this project any other comments on knowledge engineering all right in education we also had a great quarter of live streams so the two main series of live streams that we had this quarter we had some guest streams and we had our regular paper discussion live streams so in the guest streams we had uh marco facton adam saffron shannon proch anna lemke maxwell ramstead and dalton saxdevel as well as just yesterday with professor steven grossberg so a lot of awesome guest streams one off presentations and discussions on active inference research really like core and cutting edge work as well as work that might not explicitly mention active inference but is super consistent and of general interest like with professor lemke it was a great discussion on dopamine and in the paper live streams we've had paper 41 42 43 44 45 and 46 working our way through the 40s 41 was on extended active inference 42 was on robotic navigation and the slam simultaneous localization and mapping 43 was a theoretical and experimental review related to predictive coding and actin it was a mathematical review paper was therapeutic alliance as active inference the role of therapeutic touch and their ian tennant was really instrumental in coordinating those discussions and preparing as was sid in 42. in 45 we talked about free energy principle made simpler but not too simple where carl fristen and thomas parr joined for those discussions and provided some great insights into this paper and we had a very cheerful time and there was a lot of surprises and a lot of bayesian surprise too and no surprise and

amazing surprise are not the same thing as we've been discussing in the textbook and then in the last two weeks we've just completed live stream number 46 on active models do not contradict folk psychology which was great brought in some new people into live streaming had some conversations of broad interest related to folk psychology and just the way that the academic the lived and the day-to-day perspectives on cognitive science and experience all come together and even uncovered some of the adjacencies there that go into philosophical and even theological angles so awesome live streams by everyone who had joined yeah a lot of learning happening within these streams any other comments or things that people want to know particularly fun stream that they watched or joined i had so much fun in 45.1 like where i was deferred to as like the expert on the paper that's really funny like like uh like carl was like well well blue how what do you think um that was that was really like amusing and cute yeah it was really a memorable experience was um beginning 45.1 i think and when we hit start streaming blue and i were the only people in the chat and we didn't have confirmation that anyone else would join so we just trusted the plan and had authors join quickly and so just was like it was very rapid um reduction of uncertainty and yes as tradition dictates the authors um passing to us and it's that kind of like participatory engagement where even though they wrote the paper the expertise and the perspectives are still very distributed and hopefully it's striking a new and uh productive and also value aligned way to talk about research and to continue to develop research after artifacts are published in advance of other artifacts and just hope that we can continue to integrate in these discussions as like the tip of the iceberg the best of all worlds in terms of having the rigor and the clarity and the conciseness where and as we need it and also holding space everything from basic questions things that might not initially seem related everything in between so any other comments on live streams all right um we didn't talk about this in the last quarterly lab meeting but this was a paper march 1st 2022. the authors were myself uh sean applicate swanson arghan chaudhary rj corday shadee l domiti avel aka serval blue here yvonne here sid said code uh ahmet singh medemith jakob here caleb tuttle and alex here so a bunch of us and a bunch of other colleagues we had a great time working on this paper and what we did was we evaluated web 3 and blockchain technology in relationship to the emerging area of decentralized science dci we also explored some opportunities for where active inference could interface with dsi in theory and in practice as well and that highlighted some important conversations around desai in its technological aspects what kind of technology is composing or moving the frontier of desai in 2022 and then more in principle what does it mean for science to be decentralized decentralized in what way

wasn't it always decentralized what kinds of failure modes across multiple scales do decentralized systems present them themselves with so we in the first part of this paper evaluated decentralized science desai and just covered some historical threads that a lot of the co-authors just shared amazing views on we then presented an active entity ontology for science or eos and presented it in coda tables but we'll talk more about how we're continuing to develop it it is a composable and versionable modeling system that uses the active inference entity partition with the blankets and the same ontology terms that we would use to describe particular entities partitioned states like affordances and policies and so on separated types of entities into two main types active entities which are like adaptive active entities they're able to undertake actions and modify their niche and informational entities which are also partitioned entities but do not engage in action selection so like a pdf of a paper it's an informational entity it's in the epistemic niche but it's not taking actions whereas active entities ranging from humans or programs as well as organizational institutional type entities across multiple scales are engaging in communication and actions around their epistemic niche that is providing a foundation and a grounding for continued discussions that we're having about how to bring integrity and continuity amongst natural language representations visual representations and simulation type representations of complex digital and cyber physical systems so would any of the other authors like to share like anything that i missed or one part about this paper that they found useful or interesting to work on so i'll share i i thought um this paper was interesting because there was a or or there is quite quite a bit of tension between decentralization in terms of like distributed effort and labor and also like dsi as like a market based um entity or or idea and so resolving these two um these two things or like the tension and juxtaposition of of what is design is it just the distribution of knowledge and resources or is it more based on based on markets and blockchain i think that that was one of the key insights brought upon by this paper yes hopefully we'll have more spaces and times to explore and draw out some of these threads maybe even a guest stream with these authors but what is today known as web3 and blockchain is developing and growing out of and beyond financial technology perhaps there's a sense or an argument in which financial technology are some of the sinews and connective tissues by which science occurs value and financial actions occur in scientific and epistemic ecosystems is that the common denominator will we be financial reductionists with respect to how we model for example motivation and all these types of cognitive features in epistemic ecosystems at the same time there's many features of blockchain systems that make them a natural fit for scientific information it's an ongoing

discussion and and hopefully those with interest in desai or in these kinds of technologies overall we'll just stay tuned and get involved because we're just beginning this avenue of development okay any comments here or we'll continue uh next we'll talk about some research and applications we'll talk about the active block funds project so jacob do you want to provide a little summary of active block prints um so active block friends is work in progress python package that connects the recent new library for model for modeling uh agents with active inference uh called prime dp and another library called cat cat which is used a lot in economic and complex system simulations so active bug for instance right now the combination of the active imprint agents are modeled within within cad cad um and currently currently it's still very much work in progress the the overall structure of the of the package is still being developed and we're working on um kind of polishing polishing the the the implementation of prime dp and tailoring it more towards the uh cad cat style of development and also working on problems uh how to increase the complexity of the of the models with like multi-agent modeling or different constraints thanks so pi mdp just for context is a python package that has recently been developed and it provides a very simplified foothold for python programming to be implementing the kinds of markov decision processes that's mdp that are used in some formulations of and so it's an incredibly important package to use and also there are some other features that are requisite to use for systems the kinds of generative models and generative processes that we might want to implement real applied active inference settings for example parameter sweeping architectures and being able to specify execution orders and have some reproducibility around certain kinds of simulations various features that the cad cad framework provides for so as jakob mentioned at the very beginning the multi-agent case is quite interesting like does each entity perceive and then act and then we cascade through them in the same order different order or is there some other structures and execution orders how does that influence active inference simulations with metamouth i've been continuing to work on the next generation of the active infra ant colony simulation so i hope to share more specifics when we have it but this is a continuation of the work that was done in 2021 with several colleagues we're working on various other aspects of the package and this last one here connecting to ethereum or other web 3 ecosystems so just to kind of hinge on that and look into the next slide for the active block prints project there was a ecosystem support program grant available from the ethereum foundation and we proposed in a grants application about a month ago to expand our current simulations into multi-agent setting in order to explore cognition and behavior underlying desai and d5 projects the grant

was not funded so the foraging continues but it was a great learning experience and it um helped us sharpen and elaborate on everything from the substance of the package itself to the communications around it and have preliminary discussions on how do we balance in a distributed team retroactive funding proactive funding and so on specifically we were proposing to develop multi-agents some case studies and digital twins that already exist in the ethereum ecosystem porting those over into an active inference framework and helping characterize scenarios and phenomena that had previously been only approached from a top down essentially macro economic type of simulation like if the interest rate is this amount and this is the other parameter on this macro then here's how the system will evolve trying to complement that with a bottom-up cognitive entity decision making framework that could be used in a lot of interesting ways specifically we were interested about the possibility and still are of performing these kinds of cognitive audits which is a terminology used in various artificial intelligence fields but a cognitive audit could provide insights into if potentially there was some smart contract or series of cyber physical relationships where from a computer security or network security perspective it was airtight however in the cognitive niche perhaps there were destabilizing forces or dynamics and although it wouldn't have been part of our initial work we hope to eventually include cognitive features even like for example narrative and sentiment such that different kinds of information natural language reading and writing all these kinds of features of real cognitive systems could be integrated into these increasingly actionable and coherent digital twin site type simulations so if anybody is interested to get more involved in active block prints whether they have familiarity with python cad cad academ all or none of the above everybody is welcome to get involved and also if anyone has insights into how such a project might receive some funding we're also open to those thoughts anything else on the gr application we on the engagement and outreach frontier continued connecting with new and returning participants on different platforms we saw growth across our discord twitter podcast channel and youtube channel it was a good quarter of activity on these fronts and of course always feel free to share our public information or invite other people who you think might want to get involved but we continue with the work and these are our active states in the epistemic niche any funny tweets any memorable emojis um closing out the project updates on the staying on the engagement and outreach theme we have some uh directions that are coalescing relating to the engagement and outreach at active lab first we are developing an internship program that will share more details on but will provide a much more scaffolded project specific way for individuals who want to on-ramp into the lab or to step up

their activity and learn by doing that way we've written some letters of support for applications at university programs more information should these applications of our colleagues be accepted we believe that this kind of a institution to institution partnership is going to be uh there's going to be a lot of affordances for win-win opportunities with different institutions reaching out to octave lab and having a relationship where we can each contribute what we can do and be part of some joint action some generalized synchrony and also especially through the relentless work of dave douglas we've been improving our subtitles and language translation progress which we'll talk more about but language translation includes math so just at this first pass it's raising attention to the way in which whether it's audio content or textual content or morphing in between with speech to text and text-to-speech annotation and translation is the basis of rigorous scholarship as well as accessible information so making sure that whatever the initial artifact is that it can be consumed and enjoyed in the sensory modality that's most accessible to a given person and translated into the natural language computer languages and so on that are most familiar to somebody again also helps with accessibility and rigor so just like we looked at with the textbook group with the math as a language truly being translated into natural language using ontology terms that will facilitate translations amongst different languages that will just make sure that from the first step to the 99th step of learning and applying active inference that it's really a coherent and a comprehensive experience really awesome directions and if anyone sees opportunities join in engagement and outreach with us of course please feel free to just contact us any other comments on project updates on this section so the next section of this lab meeting will be about the next steps at the project scale starting with the theme of education as mentioned we are completing the first cohort first part of the active textbook group after cohort 1 part 1 we'll finish up through chapter 5 in the end of july we're going to take august off and then we are going to pick back up with two parallel cohorts there's going to be cohort one continuing on with part two of the textbook so that those first cohort can finish the textbook at least a first pass on it during 2022 we will then also begin cohort two of part one so these will be a totally new batch of people who are coming in and starting on chapter one and people from cohort one who want to just take a second pass um i think we all do in a sense whether we take the plunge to be in cohort two or not is a different question but all these materials are the kinds of things that we can just get many coats of paint on and learn and read again and again we're really excited for this second cohort because we'll be incorporating a lot of what we're learning from the first into how we proceed with this second uh cohort as well as with part two in the

first cohort hopefully this is just the first few of many many textbook group cohorts because it provides a very accessible on-ramp into learning and applying active inference the textbook is written with that kind of learning and applying in mind hopefully these textbook groups are providing a service that makes that on-ramp even smoother and more fun and with more image memes and with more connecting the dots and more exercises to test more opportunities to ask questions and i'll put the link in the youtube chat for registering to participate in an upcoming cohort does anyone else want to add anything about this next steps on the textbook group just that it's a great opportunity to get involved if you want to learn more about active inference or even if like you're starting at ground zero we've had some people come in with like no idea about the math and um yeah it's been i think really useful to work with everyone from all levels to kind of try to make this field more accessible awesome also on the education theme so as suggested in some of the updates we're learning a lot from the textbook group and other educational projects that are occurring in the lab like the active inference ontology development and live streams with education and research as our mission we've always been interested in developing some active inference courses and educational experiences alex do you want to provide some updates on how we'll be continuing here this activity on creating online course is also as many hours is uh learning by doing for our own for creating courses because we understand that for different uh audiences different courses need to be created and finally we should have like as a bunch of different ones and this introduction and introductory uh course will be an example of organizing the team of different educational roles which you understand needed to take part of creating a course which should be quality have quality and have accessibility and so on that's why based on experience and the materials and results of textbook group including and developing different types cases and examples into online form which can be supported by online platform which we'll be using from systems management school it will be the first step broad set of educational materials for any new people who will starting to learn active inference and we will create courses and improve it improve set of courses to make it accessible to as much people as possible yes in the long arc education and even just active inference this is a modified curio card yes i right click saved it and now on the top of this pile of books is the active inference textbook and it's just reflecting like one more epistemic niche modification and it is coming after of course many papers and also other kinds of educational materials that have been prepared ranging from short form workshops that have been held in different locations to blogs and github repositories like famously a barren millage of just informative links and those are all very important educational formats there is nothing

like an active inference course and for a course to be done right and versioned in our attentional situation and with our conceptual focus on active inference it's quite a challenge but there's also an amazing opportunity here for those who would like to be learning more about how to participate this is one way to be getting involved at some stage with the team or with the feedback on the initial course or of course just as with the textbook group the early cohorts who co-create in participatory way so this is super exciting and also we've had a lot of great contributions from participants dean jessica brock so many others who have added and helped us build um a lot of like a a garage of spare parts and motifs and considerations that when we formalize the course team will be able to be taken into account and implemented um all right on the education theme as well let's talk about the systems thinking summer intensive yvonne or alex wanna describe what will be happening in august 22 yes uh in august we will launch a branch activity of system thinking course and it will unpack the base concepts of the system one of the base concepts system thinking it's roles and row based descriptions for four weeks we will define what the roles are how we can use it to have a common meaning with how we can see different things a way to understand each other yeah uh that's it great alex on anything all right so in educational activities we are announcing some of the early sketching around the internship program which perhaps by any other name would it smell so sweet it's really like a learning by doing scaffolded affordance it is a self-driven opportunity for individuals of all ages larval stages backgrounds etc to engage in personal learning and development that will be related in a large part to project-based contributions to active lab so it'll be win-win you'll be learning by doing and contributing to the lab as well and helping projects develop the duration and time commitment we're figuring out some details but we'll be able to make it work for people who want to be involved we'll find a way to make it happen for those who engage in this type of internship beyond priceless value of learning and development being on the trajectory of acting inferring serving the lab can also offer everything from letters of recommendation and guidance and mentorship as well as increased access to the digital research architecture of the lab and we also hope to provide more affordances as we gain resources and organizational complexity these uh internships will be centered around a target project although there will also be curriculum of the internship that is not related to a project like related to watching live streams or engaging in the textbook group cohorts but a lot of the contribution will be focused on a project that will be carried out in a structured way so it'll provide again great opportunities for people who have skills that they would like to apply skills they'd like to learn for now the best way to learn more about this

internship whether you're interested or whether you know of somebody who might like to maybe get involved would be to contact us at activeinference@gmail.com and or to register for the systems thinking summer intensive because these projects will have role-based approaches to participating so the summer intensive is a couple weeks long a couple of hours per week and it will provide a nice warm-up for getting involved in a role-based internship program okay any other thoughts here also in our next steps expanding and scaling the active inference journal so this is again work with several participants and especially dave we've been preparing a sequence of work and practices and roles so that for our past catalog of live streams and other presentations and going forward we will be able to have a effective and automatable and also just documentable and reproducible approach preparing the transcript of a speech event into an edited curated format that can then be rendered into on one hand a prose based format which can be published for indexing citing and searching and on the other hand into a subtitle format that will facilitate accessibility of the videos and also that can all be translated across different languages so it'll be really exciting to really nail down our practices and ways of working such that within several days potentially after a live stream occurring we could publish a very high quality annotated artifact that could leverage the impact of a lot of these very important conversations that are happening anything else to add on the journal all right announce here that we are having a second applied active inference symposium on the theme of robotics it will be held on july 31st 2022 from four to eight and 16 to 20 utc so all the dates and times are utc it's a four hour first interval 12 hour break or sorry eight hour break and then um another four hour interval stay tuned on social media and our newsletter for more information including presenters we've been having a great time collaborating with co-organizers matt brown and mark miller as well as some of us on this call in the last year or two there have just been many important introductory works bridging and robotics so for example here's an image from how active inference could revolutionize robotics of decosta at all recently as well as the paper that we discussed of catal at all in april in livestream 42 on the simultaneous localization and mapping so we hope to hold this symposium where there will be presentations and roundtable type discussions that are live streamed as well as recorded and re-watchable and also with transcripts that will be published as proceedings to highlight some issues and opportunities in increasingly important area of active inference and robotics so we hope to be drawing on many perspectives those who are coming at the active robotics intersection from the perspective of researchers industries uh involved in robotics or active inference those with curiosity those with philosophical interests in these topics ethics and the

social implications there's going to be a lot of great discussions around this symposium so just stay tuned and we'll have more information coming out soon comments on the symposium or any other next steps for projects one new thing special announcement time as of june 2022 this month we are formalizing a transformation at the organizational scale from the active inference lab into the active inference institute use those emojis but this has been a really exciting development and we like to just share a little bit about it now we've been working with freed frank in a pro bono law firm capacity and they've been helping guide this developmental process we're initiating some enterprise architecture development such as updating and improving our governance structure obtaining funding and hiring for research and educational roles and identification of focal projects as affordances for onboarding more generally move in generative model or markov blanket from the active lab to the helps us clarify and continue on with our strategy around education and research and the time has come for something different and bigger and so we hope that the aii organization will be able to represent that and scaffold that and provide those services to our niche in the best possible way i guess there's a ton to share and ask and in the coming few slides we'll talk a little bit about strategy but what do people want to add here about this just for now i want to say that these changes on organizational scale from one hand reflects the growing number of services that we are providing to to the niche and from the other hand is an answer for growing complexity of activities and i believe that after we analyzing immigration projects before beginning of next semester we will be really ready for scaling of our activities and to include more people to participate and having that scale to the opportunity to grow and having as a ability to to have it is ability to grow new skills and possible groups departments and different types of projects and support different types of collaborations thanks anyone else want to add before we talk about strategy a little bit we just want to oh yeah blue please i just wanted to add that um i don't know it's a really involved process becoming a 501 3c and thinking about governance structure and how that fits into our strategy and balancing the things we have to do like obligational commitments in order to further the success of the active inference institute and also like the fun things that we like to do like research and papers and textbook groups and and things like that so so striking a balance has been like interesting challenging and hopefully it'll be rewarding it's uh only 18 months ago when we began the lab in in the beginning of and we very immediately afterwards began the process of formalizing as a non-profit we did end up doing that in a california nonprofit that was a um good step that helped us get a taste of the kinds of enterprise architecture that we would ultimately need and now we're ready to face

those challenges and opportunities really directly with this new approach so it's super exciting and we just wanted to include it in this lab meeting as recent developments and more will be coming in the coming months in July of 22 we'll take kind of a summer break there won't be any live stream paper discussions and during that time we'll be having a lot of migrations and updates so just stay tuned for things to be updating as they do in the act institute let's talk a little bit about strategy here we'll present a update on our strategy document that um as we update this at the quarterly scale we'll be helping to guide some of the next development steps in the active institute active inference institute aii or you can use emojis with just an a and then the 2i emoji is on the path of open-endedness our strategy considers learning and applying active inference for changes in the niche over multiple nested scales through time we increase the degree of hierarchical organization complexity to overcome competing interactions and frustrated states we engage in policy selection reducing our uncertainty about realizing our expectations and preferences for epistemic values as a non-profit organization to ensure institute development we need to secure pragmatic financial sustainability with enabling activities and the three scales that the aii is modifying and interacting with first the participant as an agent the human participant the institute provides affordances and updates participants generative models via providing them a modified niche with the institute as an agent this level is where we can engage in institute level policy selection and then and evolve a shared generative model that goes beyond any one participant and then lastly we are modifying and interacting with our communities and peer organizations in our epistemic niche where we provide services and opportunities and engage in collaborations with community so any overall thoughts on this our first area of focus thank you blue our first area of focus is education at the participant scale we approach the participants active inference mastery as the system of interest to reflect the plurality of individual priors expertise levels and preferences we develop different educational formats such as workshops discussion groups online courses project-based learning internships and more and our preference is to have up to a hundred participants involved in education by the end of 2022 at the institute scale we're learning state-of-the-art ways of working and practices across functional domains to build and strengthen our organizational capacities and at the community scale we work on education and reducing research debt for broad communities we develop public video and audio content that spreads and we prefer to reach out many dozens of thousands of people with the information that we create any other like thoughts or ideas on i think it's exciting to explore what the future and the presence of education are we've seen only the early days of online even though there's been

many effective or less effective approaches tried there's so much develop around how we educate especially for the active inference area two related to research at the participant scale we provide affordances to scaffold and launch research and development projects by participants projects can be driven by participants inside or outside of our organization and the institute will provide relevant support we expect to see dozens of research outcomes in the coming months and years ranging from papers workshop and conference presentations and from the basic to the applied at the institute scale as an organization we rearrange we catalyze and participate in active inference research projects with long-term impact or public goods features like ontology development knowledge engineering and also events such as symposiums and conferences at the community scale we recognize and seek to carry out apply and improve upon research practices from open science desai to be caught up education d-school and then to go beyond and create novelty and discover on the frontier research like desai those are the two key and first focuses of the active institute the third is outreach and engagement so at the participant scale we think of participants as ambassadors of the institute so we welcome and support their initiatives to share a voice about active inference in various situations and communities at the institute scale we aim to set up partnerships with other organizations from academia industry and beyond this supports high quality science communication education and collaborations around research topics such as active inference and at the community scale we acknowledge the diversity of people's initial conditions for example their familiarity with different languages science philosophy computation everything we develop projects of uh like the active inference online journal and create popular science content for different social media and we hope to provide portfolio of outreach and engagement opportunities again to meet those who are interested where and how they are any other thoughts or ideas on this so just a thought on the research for people that might have projects that didn't start in the active inference lab or projects that they want to start and haven't yet started we don't necessarily start projects sometimes projects come to us so if you're looking for a network of people or maybe weekly meetings with feedback or something else to scaffold your project we can provide access to a big community of active inference researchers and like would welcome projects that have already taken off so if you have a project that you want to bring into the lab that is open science that's i think one of the key things for for the lab um we would just welcome that opportunity to work with anyone who wants to do an active inference based project thanks blue and yeah hopefully we can have more exemplars and clear lanes and the uh mystery lanes for those that don't even yet exist but

there's a continuum of ways to participate in project-based learning ranging from coming to an act institute scaffolded project where we will help with the rhythm the infrastructure the role assignment and everything it's kind of like a show up and enter the game type project there also might be projects that are totally off the radar of act institute it's just you are participating in a project or not but you're connecting with participants in in the organization and you find a latent interest and you collaborate that way with your own font formal documents ontologies narrative and tools or something in the middle where you have a project that's at a stage where you're ready to get feedback from the active inference community or just from any interested people maybe there's opportunities to collaborate or maybe you're looking for just some help on tweaking it and fine-tuning it as you prepare for example to submit a paper or develop the next version of something so there's so many ways to do research and education that we will just have clarity around the projects that the institute is scaffolding and moving the needle on and those are the kind of massively collaborative projects where people who want to be involved can kind of get involved and check back in and out all the way to projects that are self-driven and you're seeing the lab as again a niche that is modified for you to support those kinds of research efforts like one exciting opportunity there is for individuals who would like to be writing or building applications that use active inference to gain some familiarity with the ontology could go a long way and somebody could just hang out in the relevant meetings or read the relevant documents and publications and then that is like a way to close the feedback loop and to use these services that are being provided in the niche like the ontology and so reduce research debt by promoting coherent and comprehensive use of active inference terms and concepts and so there's many ways to uh learn and research and apply aligned with this overview strategy work in progress we're presenting here and lastly four methods at the participant scale we educate participants so they grow their expertise level in different disciplines as disciplinary expertise entails competency in various practices and tools we take a holistic approach to participant professional development communication and organizational design are our fundamental effort we apply active inference systems thinking and other first principles approaches to the institute operations we share and promote methods we use throughout the community to increase the quality and efficiency of collaborations and we're also open to recommendations for new methods and strategies to consider methods is a little bit of the how just as if it were a scientific paper and the what is education research outreach and engagement this lab meeting has consisted a little bit of a introduction section again sticking with this scientific paper metaphor an introduction of updates maybe also those are

our results some discussions and next steps now with this one new thing we are also pointing the way in which we are heading during this ongoing transformation into the active inference institute it's a really exciting time those who are involved and paying attention expect and prefer there's many to be getting more involved in any number of ways ranging from educational opportunities to joining the kinds of projects where you can be engaged in research education development application many opportunities there it's just the beginning for the institute so much more will be happening as time flies what else would people like to address or ask or bring up in closing results and plans what we are having reflects our initiatives in a really long time scale and we understand that we can evaluate results like in one or two decades and save it to his first initial small steps were like a big beginning on something big as for strategy and all that we have now is uh like applications of approaches that we choose in the beginning and an example of strategy which is like a process for us and uh that updates on quarterly quarterly based updates of the strategy is important because it's provided way to be adaptive and react on changes on information we're getting in the process of work and also consideration an institute and a strategy on at least three scales of that we are paying attention to and planning changes for free scales is also providing our to us a lot of opportunities to affect aspects and niches which are related and that's why being from one hand have a very long time horizonte in our planning having but very flexible approach to like with cornerstones of first principles and kind of a evolutionary and open-endedness approach i believe will bring us very interesting result but very interesting results at the end does anyone else want to add anything or i'll make a note we can continue discussing so one one conversation that um jacob and i and others were kind of developing and and memeing out over the quarter like what if active inference only works for like one or two time steps as we consider deeper and deeper even potentially two in the most extreme versions of the meme that the combinatorics of decision making in principle and or in practice become challenging to evaluate and compute like the computational complexity grows potentially extremely rapidly considering even two moves deep so to speak we've seen several approaches to dealing with this bull first is the continuous time approximation of active inference where there is no explicit time series prediction of future time points there's just like a taylor series approximation based upon the generalized coordinates of motion other approaches we've seen are deep active inference just rolling out that prediction to deeper and deeper time points anyway and then utilizing heuristics and other algorithmic approaches to ameliorate that combo torque and we've also seen sophisticated or hierarchical active inference modeling

where potentially at two or more hierarchically nested scales of the model still only one or two time points or only a few time points have to be reflected so it'd be like to predict out a hundred years one could have a model with a temporal depth of a hundred or there could be ten time points each consisting of a decade of 10 years and so then that could even be broken down further and so when alex was talking about the way in which there's an imperative to act in an adaptive way to not die and to live and to thrive and to serve of course but in many cases there are some challenges associated with decision making we don't have the ability to do a b testing or null hypothesis or even just alternatives over very long time scales over certain levels of organization it might be possible to do for an event that's rapid or has a lot of individuals engaging with it like a website a b testing might work super well if 100 people are looking at the website every day whereas in organizational design there isn't that much of the same affordance it speaks to the way in which our multiple nested time scales of updating hopefully provide clarity so that we don't need to be at the fast cycle trying to predict out a hundred years but with increasingly deep hierarchical temporal models we might be able to act adaptively in a multi-scale way so for example in 2021 we were planning heavily on the quarter time scale introduced and assembled our scientific advisory board which began operating in and that was explicitly after one year of operations expanding our temporal horizon to the one year time scale now with the kind of architectural updates that the act institute is providing and necessitating we're thinking beyond one year explicitly because questions who can elect who in the organization they do play out over slower and slower time scales it's been quite interesting even mostly qualitatively to be applying and transferring what we're learning about multi-scale decision making under uncertainty from active inference which is kind of what it's about and apply it in the organizational multi-scale context with the participant and the projects and teams and the institute scale and the community in the niche scale and so it's already been very rewarding and we hope effective in a principled way bring our learnings and ideas about organizational design and the vast experience that so many participants do bring to the table on these issues and coordinate around carrying that out the active inference ontology in a sense and using other pieces of the puzzle that aren't now or in the future even necessarily part of the active inference ontology a lot of those features we derive from active inference isn't a project management approach so there's gonna have to be like more than just the actin phontology to carry out effective team management but that's why we use multiple ways of learning and researching so it just was very interesting how some of the theoretical discussions that we were having literally about multi-scale planning in multi-

agent active simulations and the kind of cognitive complexity that needs to happen on board the entity what is their actual cognitive complexity and awareness and affordances versus how are we going to simulate it how do we prevent the simulation from like sneaking in extra information how do we think about the role of stigma g and the modifications of the niche and the cognitive offloading that could happen in a multi-agent setting or in a stigmatic context that is in tight feedback with the ways in which we're also structuring our own niche a lot of that really started to take an acceleration point in the last several months jakob's contribution of active block friends and other individuals contributions and the seeds were planted in the work even before octave lab existed with viatgan at all 2020 on active inference behavioral engineering systems thinking and remote teams long game but then long compared to what it's just the game it is multi-scale decision making with diverse perspectives over multiple spatial and temporal scales are not like the end point or some hot take that is our starting point planning as inference i hope that all these threads can just continue to develop over the time scales that are appropriate and that we can connect with the participants who are interested for those who see that high quality engagement and relationship building worthwhile in terms of epistemic and pragmatic value sometimes that takes multiple years to approach this exciting and developing topic which is changing month by month as our guest streams just recently have been highlighting so for those who feel uncertain about participating they feel some impulse to get involved now watching the live stream or in the future yes it's easy and absolutely valid to feel busy and like it's an affordance that's not immediately preferable i'm not saying drop everything and join act institute however again we hope that those who have paid attention to what we've shared today and hopefully continue to share and for those who resonate with the approach that we're taking adaptively resonate with the strategic vision and with the increasingly clear and numerous affordances for we hope to connect with you individuals and create services that scale even beyond those who get very active in the institute itself so much more is to come again it has been quite a ride in the last um six quarters and people can check our quarterly roundtable meetings which slides are being modified up to the day basically and they're like an honest representation of how things are developing are uncertainties around uh developments in the lab we have nothing if not honesty and clarity and vision there's no clean way to end it when one has walked this deep down the plank but i'll just again you know appreciate everyone who's helped in every way to get us to this point from the people who like first were enthusiastic about the idea of an active inference lab in the beginning of 2021 and end of 2020 more than a year before the textbook when it was and still

to some extent is uh an assemblage of papers with partially overlapping focus and ontology uses and notation and amidst that uncertainty some people did step up and they acted and they inferred and they served and now we're starting another chapter the logo doesn't change everything else is just continuing to develop so it's been a great ride and thanks again to everybody who even as just a listener for now has been involved in some way really being a part of the embodiment and the inaction of active lab slash so just epistemic and pragmatic value may the gradient be with you any last comments or we can end it there thank you fellows see you later bye you